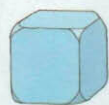


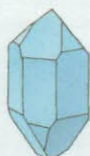
CRYSTALS AND GEMS



THE WORLD AROUND US is made of tiny crystals. Much of the Earth's surface is made of rocks that contain minerals which are, in turn, formed from crystalline particles. A crystal is a solid substance that has grown in a regular, geometric form, with a smooth, plane surface, or face, and straight edges. Crystals have many different properties, and are used in industry, and for decoration. Most gemstones are crystals prized for their beauty; they are usually cut and polished to enhance their appearance.



Cubic system contains cube-shaped crystals, and also includes 8- and 12-sided crystals.



Hexagonal and trigonal are two systems with similar symmetry.



Tetragonal systems are generally more elongated than the cube.



Triclinic is one of the least symmetrical of the crystal systems.



Orthorhombic system has prisms and flattened tabular forms as its most typical features.



Monoclinic has less symmetry than the cubic system.

Crystal systems

Well-formed crystals have regular, symmetrical shapes. The geometrical shapes in which minerals crystallize are grouped into seven crystal systems. Within each of these systems many different forms are possible, but all the forms can be related to the symmetry of that system.



Limescale crystals that form in a kettle

Crystal structure

Crystals are usually made from a single mineral. They are built up from a regular lattice, or framework, of atoms. Each atom has its own special position and is tied to others by bonding forces. The atoms of each mineral always bond together in the same way to form crystals of that mineral.

Cecil Rhodes

Known as the "King of Diamonds", British imperialist and businessman Cecil Rhodes (1853–1902) made his fortune after staking a claim in a diamond mine in Africa in 1871. By 1888 he had secured a virtual monopoly of the African diamond industry.



Cartoon of Rhodes as "King of Diamonds"

Crystal habit

Crystals form as molten magma cools, or as a liquid evaporates from a solution containing a dissolved mineral. Crystals rarely form perfectly. The conditions under which crystals grow affect their shape. The general shape of crystals is called their habit.



Prismatic crystals are prism-shaped.

Massive crystals grow in a mass in which individual crystals cannot be seen.



Dendritic crystals have tree or plant shapes.

Acicular crystals have slender, fragile, needle-like masses.



Bladed crystals look like the blade of a knife.

Aggregates are groups of crystals. This is an aggregate of prismatic crystals.



Diamond in kimberlite

Diamonds

The world's hardest substance, diamonds are sought after not only for their unique lustre, but also for use in drill bits and glass cutters.

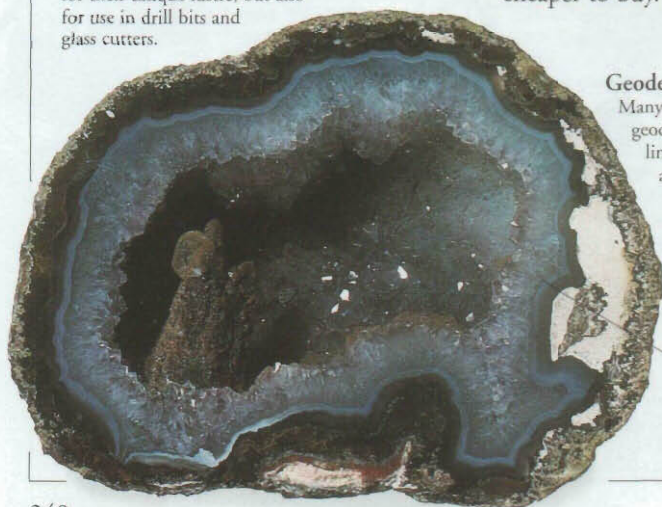
Gems

Highly valued and used in jewellery, most gemstones are beautiful, rare, and durable inorganic crystals. A few gems are organic, such as amber, jet, and pearls. Others, called synthetics, are produced in laboratories. They have similar appearances and properties to natural gemstones, but are cheaper to buy.

Geode

Many gemstones are found in geodes, round rock-hollows lined with crystals. These are formed from the bubbles of hot gas and mineral-rich fluids in magma. They are prized by collectors.

Amethyst crystals in geode



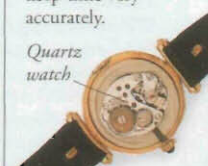
Gem cutting

Most rough crystals have to be cut and polished to remove surface imperfections and to reveal their true brilliance. Before this happens the lapidary, or cutter, studies the rough stone through a loupe (lens) to see if it is suitable for cutting.

Quartz

Quartz is "piezo-electric". This means that when an electric current passes through quartz, it resonates at such regular intervals that it can be used to keep time very accurately.

Quartz watch



1 The cutter identifies the natural grain and flaws, then marks where the diamond should be cut and ground.

2 The diamond is sawn in two to remove the top pyramid, then rounded by grinding it against another diamond.

3 The stone is mounted on a dop (stick) and several flat facets, or faces, are ground on a cast-iron wheel called a scaife.

4 Slanted side facets called bezels are ground between the table facet (the flat top) and the girdle (the rim).

5 The stone is turned over and faceted in the same way. It is then finished by a brillianteer who adds 40 small facets.



FIND OUT MORE

ATOMS AND MOLECULES

CLOTHES AND FASHION

ROCKS AND MINERALS

SOLIDS

TIME